

SmartCare v0.1 - Initial Engineering Brief and AI Activity Card

A2 Case Study Stage 1 student resource

SmartCare scenario

SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The client says: 'We need software to help manage patients, practitioners and appointments.' This is not yet a complete specification.

Initial Engineering Brief

1. Problem summary

SmartCare Community Clinic relies on paper records and spreadsheets to handle patient details and bookings. This manual setup causes issues like lost files, duplicate entries, slow and long waiting times and even potential privacy risks. To fix this, the clinic needs a smooth and straight forward software system that focuses on patient records, doctor schedules, and appointment management tools. Moving to a digital platform will cut down on human error, keep patient data secure, make daily operations run smooth and quick, and an overall shorter waiting times for not only patients but for everyone.

2. Initial stakeholders

Stakeholder	Possible need
Patients	Quick booking options and shorter waiting times
Receptionists	A simple system to book appointments without overlapping times
Doctors	Quick and reliable access to full patient records during appointments
Clinic management	Proper data security compliance and basic reporting/tracking on clinic operations

3. Initial features

Feature	Confirmed or provisional?	Why?
Patient record management	Confirmed	Requested by the client to replace paper forms
Appointment scheduling	Confirmed	Directly requested by the client to manage bookings
Practitioner schedule view	Confirmed	Essential for tracking doctor availability as mentioned
Online billing and payment	Provisional	Not mentioned in the brief, so we need to mention it to the client

4. Questions for the client

1. What different permission levels should admin staff have compared to doctors?
2. How many staff members and doctors will be using the system at the same time?
3. What specific patient details and medical history fields do we need to store?
4. Are there any specific healthcare privacy laws or regulations we need to consider?
5. Does the software need to connect with any existing external systems, like billing software etc.?

5. What we do not yet know

1. The exact computers or hardware setup currently available at the clinic.
2. The detailed workflow for how staff could handle cancellations or rebooking.
3. The specific budgeting factors that the clinic may have and when the expected deadline/launch.

AI Activity Card - Ask, Check, Explain

Before AI

- What do I think the code does?
I think that the code is meant to take in patient names and save appointment dates into the software's memory but only temporarily.
- What problems can I already identify?
The problems I can already identify is that it doesn't validate user input, it allows double booking the same time slot, and data resets every time the app restarts.

AI request

Act as a tutor. Explain this code and identify potential problems. Do not provide a complete replacement. Ask me questions that help me reason about the solution.

Evaluate

Suggestion	Useful	Unclear	Incorrect	Out of scope
Add input checks so users can't input incorrect dates	Yes			
Connect to a database so the data stays saved	Yes			
Create everything into a full on specked out software system				Yes
Add a real time live video consultation aspect				Yes

Decide

- "Add input checks so users can't input incorrect dates"
Accept adding input validation so invalid dates don't break the app.
- "Connect to a database so the data stays saved"
Accept the database idea to use simple local database to keep things straightforward.
- "Create everything into a full on specked out software system"
Reject creating every feature and aspect into a fully specked and high-end software system.
- "Add a real time live video consultation aspect"
Reject adding a live video meeting/consultation feature.

Verify

- Run the code – to make sure there are no syntax errors.
- Test normal input – by typing in standard patient names and valid dates.
- Test unusual input – through entering letters into numerical fields such as in the date fields or booking a slot that's already taken.
- Compare with requirements – to check if all basic appointment features are working fine.
- Ask tutor/peer – to check if my logic for checking double bookings makes sense.
- Check documentation – for automatic date formatting functions in Python.

Explain

- Can I explain the final code without reading the AI response?
I can somewhat explain the final code without reading AI's response, the code checks if the entered date and time are valid and is open before adding the appointment confirmation to our database.
- What do I still need to understand?
How to encrypt passwords properly and manage user login permissions safely.